

Project Proposal: Agentic Wine Recommendation System

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Problem Statement & Motivation

1. What real-world business problem are you addressing?

Casual wine consumers often have a hard time finding high-quality wines that meet their taste preferences. Gift buyers may have a hard time finding high-quality wines on a limited budget. Additionally, due to the numerous different types of wine available, navigating all of them can be tedious. Users need a way to find wines that meet their taste preferences, meet their budget, and filter out low-quality options instantly without reading dozens of reviews themselves.

2. Who are the target users?

- a. The Average Wine Consumer: Users who want wines that fit their taste preferences and have high ratings.
- b. Gift Buyers: Users who want to gift high-quality wines with a limited budget.

3. Why is an agentic AI solution appropriate?

A single search often returns unhelpful or mixed results. An agentic system can help the user tremendously by checking wine details and automatically verifying metrics. It can check taste details and ratings across multiple sources before a recommendation is ever shown to the user.

Use Cases & Scenarios

1. Scenario 1: Rating & Price Check (Gift Buyer)

- a. **User Intent:** Find me a highly-rated Pinot Noir under \$20.
- b. **Agent Tasks:**
 - **Researcher Agent:** Searches online retailers for "Pinot Noir under \$20" and finds 10 candidates.
 - **Validation Agent:** Scans the results for rating numbers. It flags and removes bottles that are rated lower than 4.0 or 80%.
 - **Advisor Agent:** Presents the remaining verified options to the user.

2. Scenario 2: The Taste Search (The Average Wine Consumer)

- a. **User Intent:** Find me a full-bodied wine that is oak-forward. It should be between \$30-\$50.
- b. **Agent Tasks:**
 - **Researcher:** Searches for oak-forward full-bodied wine under \$30.
 - **Validation Agent:** Scans and rejects any wine with less than 4.0 stars or 80% rating. It also checks the reviews to confirm if users actually mention the taste preferences (oak).
 - **Advisor:** Presents the top wines that met the taste preference and rating checks.

Initial System Design

1. **General Info:** This system uses a multi-agent workflow (Researcher → Validation → Advisor) to ensure quality control.
2. **Preliminary List of Agents and Roles:**

- a. **Researcher:** Searches the web for wines matching the user's general criteria (price, region, grape, taste, etc.).
- b. **Validation Agent:** It accepts the list from the Researcher and strictly filters out any item with a rating below 4.0 or 80%. If a wine has no rating data, it is discarded to be safe.
- c. **Advisor:** Takes the list from the Validation Agent and formats it into a friendly recommendation for the user.

3. Expected Tools/APIs (still deciding exact tools):

- a. **Search Tool:** To fetch product pages and review summaries.
- b. **System (like Google Opal):** To pass data from Researcher → Validator → Advisor.
- c. **LLM (like ChatGPT):** To give recommendations in a user-friendly manner.

Feasibility & Risks

1. Potential Challenges:

- a. **Taste Differences:** One review might say the wine is very sweet while others may say it is semi-sweet.
 - Mitigation: The Validator agent will look at the information in the wine description/details and check reviews.
- b. **Empty Data:** Some wines might not have reviews.
 - Solution: The Validator will reject wines with missing details or reviews/ratings to prioritize user confidence.

2. Timeline:

- a. **Week 4:** Proposal Submission.
- b. **Week 8 (Milestone I):** Front-end UI.
- c. **Week 9 (Milestone II):** Build a multi-agent workflow (Researcher, Validation Agent, and Advisor).
- d. **Week 10 (Milestone III):** Evaluate if the system successfully recommends wine that meets parameters (finds wine with correct taste preferences and blocks low-rated wines in 10 test cases).